IC next-10 item 3: PKM tower solving degree

2026-10-14

Item 3: the PKM tower's solving degree past the memory wall

Verdict: NEGATIVE for the `m = 4` plateau, INCONCLUSIVE for `m = 3`, and the `d + δ` conflict is resolved.

- **The plateau at 7 does not hold at `m = 4`.** At `m = 4`, `N = 20`, on **both targets**, msolve 0.6.5 completes **three productive degree-8 F4 rounds**, adding 1,241, 2,507 and 3,578 basis elements. It then hits the 5 GB cap. The two traces are identical.
  - msolve takes pairs of the lowest degree first, so it reaches degree 8 only after every pair of degree ≤ 7 has been reduced. The degree-7 closure therefore does not contain `1`, and **`D ≥ 8` at `m = 4, N = 20`**.
  - The `m = 4` line now reads `D = 6, 7, 7, ≥ 8` at `N = 8, 12, 16, 20`. The first two are msolve's, matching `f4_fp_tower`. The value at 16 is `f4_fp_tower`'s; msolve reached 49 rounds there with no degree above 7 before the memory cap.
  - Under the note's rule for a stopped run ("a productive step at degree 8 or more"), this reads **H0: growth resumes**. It is about one degree per eight tower unknowns over `N = 12`–`20`.
  - **Caveat:** the `D ≥ 8` is msolve's, on the raw presentation. `f4_fp_tower` (reduced presentation) has not run `N = 20`, which needs more than 14 GB. The two engines agree on every one of the 16 control systems both finished.
- **`m = 3` stays inconclusive.** At `N = 18`, 21 and 24 msolve completes one productive degree-7 step (`D ≥ 7`). It is stopped by the 45-minute wall inside the second degree-7 step, at 2.6 GB, before any degree-8 round.
  - The design's kill test, `D ≥ 8` at `N = 21` and `≥ 9` at `N = 24`, can be evaluated neither way.
- **Positive control: passed.** msolve matches the committed `D` and verdict on all 16 control systems it finished:
  - `m = 2`, `N = 12`–`18`: `D = 5` (8 of 8);
  - `m = 3`, `N = 9, 12`: `D = 6` (4 of 4);
  - `m = 4`, `N = 8`: `D = 6`; `N = 12`: `D = 7` (4 of 4).
  - At `m = 3`, `N = 15` it reaches a productive degree-7 step (`D ≥ 7`, committed 7) before the time cap.
- **Where the wall falls.**
  - **`m = 4`: memory.** All three `m = 4` jobs ended with a segmentation fault (exit 139) on allocation, at `VmPeak` 4.86–4.87 GB of the 5 GB cap (resident 3.7 GB):
    - `N = 16` at round 50, a degree-7 round with 31,902 pairs;

        - `N = 20` at round 22, the fourth degree-8 round, with 13,262 pairs, on both targets.
    - **`m = 3`: time.** The host ran at a load of 22–25 on 4 cores, so a single-threaded job got about 19% of a core. `N = 15`, 18 and 21 were stopped at 45 minutes (`N = 24` by the operator at 27 minutes), each inside the second degree-7 step. Their peak resident memory was 1.5–2.6 GB.
- **The `d + δ` conflict is resolved.**
    - For the `m = 2` full presentation, Proposition 2 is an equality. `δ = N` exactly at `N = 12`–`26`: the inverse `1/S̄` has every one of its `2^N` square-free coefficients non-zero, and its weight histogram is exactly `C(N, k)`. So the Macaulay/XL degree is `D_XL = N + 4`, from 16 at `N = 12` to 30 at `N = 26`, while F4 needs 5–6.
    - A direct Macaulay rank computation confirms `D_XL = 16` at `N = 12`. It gives `D_XL = 13` at `m = 3`, `N = 9` (F4: 6), `≥ 14` at `m = 3`, `N = 12` (F4: 6), and 12 at `m = 4`, `N = 8` (F4: 6).
    - The bound is real and tight, but it bounds XL only, which never reuses a reduced row at a lower degree. F4 escapes it through degree falls.
    - So the conflict is not evidence for a plateau. The `m = 4`, `N = 20` result shows that F4's degree does rise, far more slowly than `N`.
    - The Hilbert-series predictors fail as the note's §3.4 says they must:
        - the top-part ideal of the full presentation has series `(1+t)^N`, so `d_reg = N + 1`;
        - the chain presentation is positive-dimensional at infinity and has no `d_reg`;
        - the homogenised generators' affine Hilbert function reaches 0 only at `D_XL`.

      None predicts `D` at any `N ≤ 18`, so none is used beyond.
- **A structural observation: level windows.** msolve's F4 trace at `m = 3` is identical round for round at `N = 18`, 21 and 24 through the first degree-7 step: selected pairs, rows, new elements and zero reductions. At `N = 21` and 24 the columns match too. That step adds 2,354 elements at every one of these sizes, and so did `f4_fp_tower`'s step 19 at `N = 18`.
    - At `m = 4`, the two `N = 20` targets have identical traces.
    - F4's work up to the first top-degree step therefore does not depend on `t` once `t` is large enough. What grows with `N` is the phase after it:
        - at `m = 3`, two degree-7 steps at `N = 15` and six at `N = 18`;
        - at `m = 4`, 38 degree-7 steps at `N = 16`, against 5 degree-7 rounds followed by degree 8 at `N = 20`.

1. Setup

- **Systems.** These are the Kummer tower systems at `p₁ = 2013265921 = 15·2^27 + 1`, chain presentation for `m ≥ 3`, in the raw presentation (as written, not reduced modulo the tower).
    - The committed systems are from `research/magma_f4_comparison_20261013/scripts/`.
    - The new sizes were drawn by `examples/pkm_tower_pilot.rs` with round 2/7's flags (`--random 2 --planted 0`, same seed). The run was stopped at once with `--max-nnz 1000`, and the systems were exported with `--export-magma-rows`.
    - **Identity check:** the draw at `N = 18` (`m = 3`) and `N = 16` (`m = 4`) gives the committed targets. The polynomials are byte-identical to the committed exports; only the comment line with the row's own verdict differs.
- **Conversion and analysis.** `examples/ic_next10_03_pkm_degree.rs` (new; Rust) does four jobs:
    - `convert` turns Magma into msolve's format, keeping the variable order;
    - `delta` computes `δ` by a 2-D number-theoretic transform on `(Z/2^t)^2`;

    - `macaulay` computes the exact XL degree and affine Hilbert function in the tower quotient. No row is ever reused at a lower degree;
    - `top` computes the top-degree parts modulo y^2, and the Hilbert function of that ideal when it is monomial.
- **msolve.** Every run used `msolve -t 1 -v 2`, `nice`, `prlimit --as=5e9` and `timeout 2700`.
    - `run.sh` samples peak `VmHWM`/`VmPeak` from `/proc` every second.
    - `parse.sh` reads the `-v 2` round table. `D_msolve` is the largest round degree. `D_last_productive` is the largest degree of a completed round that added basis elements; this is the lower bound for a stopped run (note §16.1).

2. Results

2.1 msolve against the committed F4 degree

m	N	targets	committed `D_f4` (`f4_fp_tower`)	msolve	verdict match	peak RSS	wall (s)
2	12	2	5	5, 5	yes (`[-1]`)	6 MB	1
2	14	2	5	5, 5	yes	6 MB	1
2	16	2	5	5, 5	yes	43 MB	6–7
2	18	2	5	5, 5	yes	171 MB	45–116
2	20	1	6	stopped at 45 min in a degree-6 step (`D ≥ 5` productive; degree 6 reached)	—	681 MB	2701
3	9	2	6	6, 6	yes	6 MB	1
3	12	2	6	6, 6	yes	100 MB	36
3	15	1	7	stopped at 45 min in the 2nd degree-7 step: `D ≥ 7`	—	1.47 GB	2701
4	8	2	6	6, 6	yes	6 MB	2–3
4	12	2	7	7, 7	yes	477 MB	595–1242
4	16	1	7	5 GB cap at round 50, 37 degree-7 rounds, none above 7: `D ≥ 7`	—	3.76 GB	2327

Every finished control agrees in `D` and in verdict: 16 systems, with 16 `[-1]` (no solution) verdicts. msolve's sequence of round degrees has the same shape as `f4_fp_tower`'s: it climbs to the plateau degree, then falls and finds `1` at a lower degree.

Instrument check at p_1. The note warns that msolve 0.6.5 failed a known-answer battery above `2^16` in another project. On these systems, at $p_1 \approx$ `2^31`, all 16 verdicts agree with the committed rows, and those rows were verified against exhaustive search. The linear-algebra variants `-l 2` (default), `-l 42` and `-l 44` give identical round tables at `m = 3`, `N = 12` (`runs_l42/`).

2.2 The new sizes (and `N = 16` at `m = 4`)

m	N	target	how it ended	rounds	degree sequence (run lengths)	D lower bound	largest matrix	peak RSS	VmPeak
3	18	0	45-min wall, 2nd degree-7 step	23	$4^3\ 5^4\ 6\ 5^3\ 6^8\ 7\ 6^2\ 7$	7	170,436 × 184,453	2.58 GB	3.66 GB
3	21	0	45-min wall, 2nd degree-7 step	22	same as 18 through round 20	7	170,445 × 194,842	2.58 GB	3.66 GB
3	24	0	stopped by the operator at 27 min (memory budget), 2nd degree-7 step	22	same as 21 through round 21	7	170,445 × 194,887	2.59 GB	3.67 GB
4	16	0	**5 GB cap** (segfault on allocation), round 50	49	$4^3\ 5^5\ 6^4\ 7^{37}$	7	118,846 × 167,176	3.76 GB	4.86 GB
4	20	0	**5 GB cap**, round 22 (4th degree-8 round)	21	$4^3\ 5^4\ 6^4\ 7^5\ 8\ 7\ 8\ 7\ 8$	**8**	213,684 × 320,433	3.71 GB	4.87 GB
4	20	1	**5 GB cap**, round 22	21	identical to target 0	**8**	213,684 × 320,433	3.70 GB	4.86 GB

- **The degree-8 rounds at `m = 4`, `N = 20`.** Each round's `(sel, pairs, rows × cols, new, zero)` is `(4432, 10966, 78739 × 134970, 1241, 3193)`, `(9955, 25599, 151098 × 236724, 2507, 7448)` and `(13412, 51333, 213684 × 320433, 3578, 9868)`. Each is followed by a short degree-7 round on the new elements' pairs.
- **The committed `f4_fp_tower` run at `m = 4`, `N = 16`** has degree run lengths $4^2\ 5^5\ 6^4\ 7^{38}\ 6^4$, and refutes at 7. msolve at `N = 16` followed the same shape for 49 rounds (37 degree-7 rounds) before the cap.
- **Timing at `m = 3`.** The first degree-7 step took 2,024 s wall and 384 s CPU at `N = 18`, and 2,041 s wall and 396 s CPU at `N = 21`. The committed `f4_fp_tower` run at `N = 18` needed six degree-7 steps and 176 minutes on four unloaded threads.

2.3 The trace is independent of `t` up to the first degree-7 step

The `(deg, sel, pairs, rows, cols, new, zero)` rows of `logs/trace_*.txt`:

- `N = 18` and `N = 21` agree on rounds 1–19 in every field, `cols` included. In round 20, the first degree-7 step, they agree in every field but `cols` (101,390 against 101,435), and both add 2,354 elements with 18,321 zero reductions. They diverge after that.
- `N = 21` and `N = 24` agree on all 21 rounds **in every field, `cols` included**.
- `N = 15` agrees with them in every field through round 7. From round 8 on its `sel`, `pairs` and `new` stay within a few units, while `rows` and `cols` are smaller: 7% fewer rows and 19% fewer columns in the first degree-7 step. That step adds 2,352 elements against 2,354.

2.4 XL degree, δ, and the Hilbert series

m	N	D_XL (method)	D_F4	gap
2	12	16 (NTT δ = 12; Macaulay rank: 16)	5	11
2	14	18	5	13
2	16	20	5	15
2	18	22	5	17
2	20	24	6	18
2	22	26	6	20
2	24	28	—	—
2	26	30	—	—
3	9	13 (Macaulay rank, both targets)	6	7
3	12	≥ 14 (Macaulay rank; $1 \notin$ span at D = 13, 40-min cap)	6	≥ 8
4	8	12 (Macaulay rank, both targets)	6	6

- **m = 2.** On every target $\bar{S}$ has no zero on the grid (σ = 0). The coefficients of $1/\bar{S}$, counted by weight, are exactly C(N, k) for every k, so the inverse is fully dense and δ = N.
- **Proposition 2 is an equality here.** NF(I_(D)) = $\bar{S}$·R_{≤D−4}, so D_XL = 4 + δ = N + 4 exactly. The direct Macaulay rank at N = 12 reproduces it independently: 1 enters at D = 16.
- **The gap grows by one per unit of N** at m = 2, from 11 to 20. This supports H-XL as the protocol fixed it.
- **The affine Hilbert function of the homogenised generators** at m = 2, N = 12 is #sqfree(≤D) − #sqfree(≤D−4) up to D = 15, then 0 at 16 (logs/macaulay_*m2_t6*.json). At m = 3, N = 9 it is still 577 when 1 enters at D = 13.
- **The top-part (degree-of-regularity) series** (top):
  - m = 2: top(S) ∈ (y^2), so the series is (1+t)^12 and d_reg = 13 at N = 12;
  - m = 3: the surviving top form is c·y0_0·y1_0·$u0^2$. The Hilbert function stabilises at 384, the ideal has dimension 1 at infinity, and there is no d_reg;
  - m = 4: the top ideal is not monomial (it has $u0^2u1^2$ + …) and is also positive-dimensional.

  No Hilbert-series quantity reproduces D_F4 at any N ≤ 18. By the protocol's rule, none is trusted beyond.

How this resolves the conflict. - The d + δ bound concerns the Macaulay space I_(D): combinations Σ h_k f_k with every deg(h_k f_k) ≤ D. - F4 at step degree D builds rows m·g with deg(m·g) ≤ D, where g can be a *previously reduced* element whose representation in the f_k has a much higher degree (a degree fall). - For Kummer towers the falls are everywhere: y_j·y_j = y_{j+1} is a carry, and §2.3 shows F4 re-deriving the same low window of the tower at every t. F4's degree is therefore bounded by the *last fall degree*, not by d + δ. - The two numbers measure different algorithms. Their gap is now measured (11 → 20 at m = 2), and the conflict is not evidence against the plateau. - It is not evidence for the plateau either. A fall-based argument still has to show that the degree-7 phase closes without degree 8 as t grows, and that is what §2.2's wall left open.

3. Where the wall falls

This host gave each job a 5 GB address-space cap and 45 minutes of wall time, and ran at a load of 22–25 on 4 cores, about 19% of a core per job.

- **`m = 4`: the memory wall** sits between `N = 12`, which finishes at 0.48 GB resident and 1.54 GB address space, and `N = 16`.
  - `N = 16` ran out of the 5 GB address space at round 50, a degree-7 round with 31,902 pairs, after 37 degree-7 rounds.
  - `N = 20` ran out at round 22, the fourth degree-8 round with 13,262 pairs, on both targets.
  - Every failure was a segmentation fault on allocation: `VmPeak` reached 4.86–4.87 GB of 5.0 GB, and resident memory 3.7 GB.
- **`m = 3`: the time wall** sits between `N = 12`, which finishes in 36 s, and `N = 15`, stopped at 45 minutes.
  - Every run at `N = 15`–`24` was inside its second degree-7 step when it was stopped, at 1.5–2.6 GB resident.
  - At `m = 3` the memory wall was not reached, so where it falls is not known.
- **For comparison, the previous wall.** `f4_fp_tower` needed 13 GB at `m = 3`, `N = 18`, and 8.1 GB at `m = 4`, `N = 16`.

4. Deviations from the protocol

1. **The prime.** $p_1 \approx 2^{31}$ is used, not the design text's "$p \approx 2^{24}$", because the plateau was measured at p_1. This was declared in `PROTOCOL.md`.
2. **Controls dropped.** Two control replicates were started and then aborted by the operator to free the job slots: `m = 2`, `N = 20` target 1 and `m = 3`, `N = 15` target 1. They are kept as `logs/aborted_*.log` and `progress.txt` records them; neither is used. The `m = 2`, `N = 22` controls were never started.
3. **Concurrency.** Up to four jobs of this item ran at once, each under its own 5 GB cap. At about 19:53 UTC their summed resident memory reached about 6.9 GB, and the host had 2.5 GB available. The `N = 24` run was stopped (exit 143) to bring the total down. Its trace up to that point is kept and used only as a lower bound.
4. **Kill criterion at `m = 4`.** The design states its kill criterion for `m = 3`. The `m = 4` reading (H0 from a productive degree-8 step) applies the note's §16.1 rule for stopped runs to the `m = 4` line. That line mixes engines: `N = 16` is `f4_fp_tower`'s, and `N = 20` is msolve's.
5. **Arm PL (planted systems) was not run.** The instrument check rests on 16 agreeing verdicts and degrees instead.
6. **Exhaustive verification.** None was needed for new verdicts, because no new system finished.

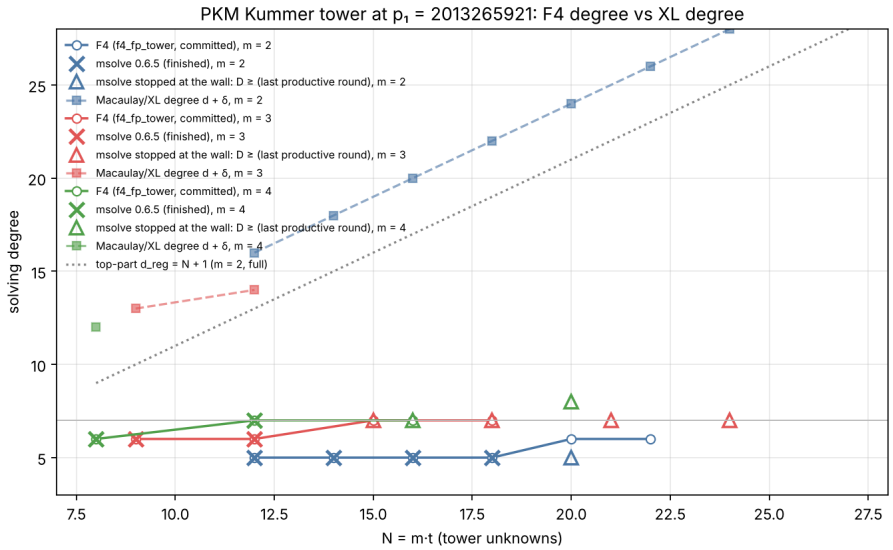
5. Reproduce

```
        cargo build --release --example pkm_tower_pilot --example
ic_next10_03_pkm_degree
        cd research/ic_next10_20261014/03-pkm-degree
        B=../../../target/release/examples/pkm_tower_pilot
        X=../../../target/release/examples/ic_next10_03_pkm_degree
        # draw the new sizes (stopped at once) and export them
        $B --engine tower --p 2013265921 --kinds kummer --controls
tower --planted 0 --random 2 \
           --ladder-t none --budget 60 --max-nnz 1000 --m 3 --t-min 7
```

```
--max-t-m3 7 --out gen/m3_t7.jsonl
        $B --export-magma-rows gen/m3_t7.jsonl --export-dir gen/magma
        $X convert
gen/magma/pkm_kummer_p2013265921_m3_t7_x221026520.magma
systems/pkm_kummer_p2013265921_m3_t7_x221026520.ms
        ./run.sh pkm_kummer_p2013265921_m3_t7_x221026520       #
msolve -t 1 -v 2, 5 GB, 45 min
        ./parse.sh > results.csv
        $X delta
gen/magma/pkm_kummer_p2013265921_m2_t12_x1002906059.magma    # D_XL =
N + 4
        $X macaulay
../../magma_f4_comparison_20261013/scripts/pkm_kummer_p2013265921_m3_t3_x388393539.ma
40
        $X top
../../magma_f4_comparison_20261013/scripts/pkm_kummer_p2013265921_m3_t3_x388393539.ma
        python3 -I plot.py
```

6. Files

- `PROTOCOL.md`: the protocol, frozen before any data.
- `results.csv`: one row per msolve run.
- `xl.csv`: `D_XL` per system.
- `f4_committed.csv`: the committed `f4_fp_tower` degrees.
- `run.sh`, `parse.sh`, `plot.py`.
- `progress.txt`: the run log, with exits; `124` is the 45-minute cap.
- `runs/`: msolve's `-v 2` logs, outputs and memory samples.
- `runs_l42/`: the linear-algebra-variant check.
- `logs/`: traces, δ (`delta_m2.jsonl`), Macaulay and top-part output, and the conversion index.
- `systems/`: msolve inputs.
- `gen/`: the stopped draws and their Magma exports.
- `degree_vs_N.svg` and `degree_vs_N.png`: the graph.
- `README.pdf`: this report.



Solving degree against N